

TOLEDO LIGHT 2, OH, USA

WMO: 997344

Lat: 41.826N

Lon: 83.194W

Elev: 188

StdP: 99.09

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-13.4	N/A	N/A	N/A	N/A	N/A	N/A	18.0	-11.9	16.5	-6.6	7.1	280	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	29.4	N/A	28.2	N/A	27.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.8	14.3	12.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-3.7	-3.2	1.3	8.6	15.2	20.9	23.5	23.3	19.6	12.7	5.8	1.3
	DBStd	10.52	5.49	5.59	4.44	3.81	3.91	3.08	2.64	2.29	3.41	4.26	4.63	4.06
	HDD10.0	1563	423	369	270	70	3	0	0	0	0	19	137	271
	HDD18.3	3337	682	602	528	291	112	11	1	1	25	182	375	528
	CDD10.0	1758	0	0	1	29	166	328	419	411	289	103	12	1
	CDD18.3	492	0	0	0	1	16	88	162	153	64	7	0	0
	CDH23.3	2202	0	0	0	1	43	317	923	712	193	13	0	0
	CDH26.7	312	0	0	0	0	4	43	158	78	27	2	0	0
Wind	WSAvg	6.7	7.9	7.1	6.6	7.4	6.1	5.7	5.4	5.4	6.2	7.4	7.6	7.3
Precipitation	PrecAvg	847	50	48	61	83	86	88	83	85	80	61	65	59
	PrecMax	1192	91	130	140	128	147	203	163	200	158	156	167	153
	PrecMin	642	14	11	14	29	20	43	32	16	14	12	20	16
	PrecStd	125	22	29	28	28	41	35	29	39	36	34	30	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.9	14.0	22.3	26.9	29.9	30.9	30.1	29.4	25.1	17.5	12.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	8.0	7.9	11.0	19.5	24.7	28.0	29.7	28.6	27.0	22.5	15.3	10.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	5.7	5.2	8.6	16.8	22.7	26.5	28.6	27.5	25.5	20.6	13.3	8.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	3.6	3.2	6.3	14.4	20.9	25.2	27.4	26.5	24.2	18.7	11.9	6.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	5.6	5.8	5.5	5.1	4.9	4.8	5.0	4.8	4.6	4.0
		MCDBR	6.6	7.5	8.1	10.4	7.6	6.6	5.9	5.6	6.5	5.6	6.7	6.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.304	0.335	0.357	0.401	0.436	0.449	0.441	0.433	0.388	0.362	0.322	0.316	
	taud	2.410	2.301	2.328	2.239	2.187	2.201	2.245	2.251	2.354	2.407	2.481	2.408	
	Ebn at Noon	843	862	887	868	843	830	833	828	844	824	817	792	
	Edh at Noon	79	103	113	133	144	143	135	130	110	92	74	74	
All-Sky Solar Radiation	RadAvg	1.49	2.37	3.48	4.62	5.42	6.06	6.08	5.32	4.30	2.82	1.78	1.23	
	RadStd	0.15	0.31	0.32	0.39	0.46	0.40	0.33	0.20	0.36	0.23	0.24	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page